Seclusion EDA Findings – CRDC 2015-16

Prepared for: Adhya Putta · October Milestone · Scope: 2015-16 school-level data, with a 2015-16 vs 2017-18 comparison

Bottom line

2015-16 looks like 2017-18: seclusion is rare (5.5% of schools report any), heavily skewed, and driven by state reporting context. The same target definition works for both years: a **binary any-seclusion target** on schools with at least 30 students, with an **elevated tier** as a sensitivity target, and models run **with and without state**. Physical restraint counts are a strong leak and must be excluded. Race could not be assessed because it is not in the cleaned file.

Data completeness

- All 11 audit counts in the cleaning script reproduce exactly (96,360 schools, 93,966 with a valid outcome, 608 justice facilities, and so on).
- Seclusion is missing for 2,394 schools (2.5%, versus 6.8% in 2017-18): 1,806 structural skips (-9) and 588 missing (-5). Dropped schools are small (median 161 students vs 449) and more often special-education (11.6% vs 2.0%), so the analysis set under-represents high-risk schools.
- The IDEA enrollment count is suppressed for 10.6% of schools. This is informative: those schools are small and rarely report seclusion (about 2.5% vs 5.8%).
- Law-enforcement FTE is 73.6% missing and unusable. There is no virtual-school indicator.
- The file is named analysis-ready but has no readiness flag, so the subset must be derived.

What the data show

- **Distribution:** 94.5% of schools report zero. Among the 5,165 that report any, the median is 4 instances, the 90th percentile 50, the maximum 3,088. The top 100 schools hold 39.8% of all instances.

- **State dominates:** Wisconsin, Maine, Iowa, Connecticut, Vermont and Kansas have 16-28% of schools reporting any; California, Texas, Nevada and New York are around 1% or lower; Hawaii reports none. State alone gives ROC-AUC 0.79, and adding it to a baseline model lifts average precision from 0.12 to 0.25.
- **Concentration:** 12.2% of districts have any school reporting seclusion; 70 districts hold half of all instances.
- **School type:** special-education schools 15.2% any (vs 5.3%) and 21.8% of instances; justice facilities 6.7%; alternative 2.5%, charter 2.7%, magnet 3.8%. Elementary 7.4%, middle 4.3%, high 2.5%. 93% of positive schools are ordinary schools.
- **Population:** %IDEA is the clearest positive correlate (2.0% any under 5% IDEA, 6.7% at 15-20%). %504 is negative at high levels; %EL is inverted-U. All rank correlations are weak ($|\rho| \leq 0.05$).
- **Staffing:** non-monotonic and confounded. After adjustment, psychologists and social workers are slightly positive and security guards negative; counselors and nurses add little.
- **Small schools distort rates:** schools under 30 students are 3.4% of schools but 30% of those with more than 100 instances per 100 students.

Across years (2015-16 vs 2017-18)

- 86,019 schools appear in both. Prevalence rose from 5.65% to 6.74% in the matched set.
- Seclusion persists: 47.2% of schools with any in 2015-16 also have any in 2017-18, versus 4.3% of those with none. The same holds for the elevated tier (47.2%).
- State-level prevalence is almost identical across years ($r = 0.97$), so state differences are structural, not year-to-year noise.

Equity (sex and disability)

- Boys are 77% of secluded students.
- Students with IDEA are 12.7% of enrolled students but 68.6% of secluded students: 0.40 secluded students per 100 vs 0.026 for others (about 15x). IDEA boys are secluded at 0.475 per 100, about 31x the rate of non-IDEA girls.
- Secluded IDEA students were secluded 4.1 times on average vs 1.9 for others.
- Race could not be assessed.

Leakage

- Physical restraint instances: 26.1% of schools with restraint report seclusion vs 1.4% without (AUC 0.84 alone). Adding restraint to the baseline raises ROC-AUC from 0.69 to 0.86 and average precision from 0.12 to 0.38. Mechanical restraint is rare (1.3%) but belongs to the same module. Exclude both.
- The three seclusion component counts, the rate and the secluded-student counts are the outcome itself.

Recommended target definition

- **Primary,** `any_seclusion`**:** seclusion_instances ≥ 1. Prevalence 5.6% in the modeling frame.
- **Secondary,** `elevated_seclusion`**:** seclusion_instances ≥ 5 and rate ≥ 1 per 100 students. Prevalence 2.5% (2,296 schools).
- **Modeling frame:** valid outcome, enrollment ≥ 30 (drops 3,212 schools), excluding FTE-outlier rows, with cross-validation grouped by district. Keep schools with suppressed IDEA counts and add an indicator. For rate work, add shrinkage (about 30-100 pseudo-students).
- **Not recommended:** raw rate or count as a regression target.

Candidate feature shortlist

- **Include:** log enrollment, grade band, special-education / alternative / charter / justice flags, %IDEA plus an IDEA-suppressed indicator, %504, %EL, psychologists and social workers per 100 students, log district enrollment.
- **Optional:** magnet flag, counselors, nurses and teachers per 100, security guards per 100 (reverse-causality caution).
- **Second model only:** state (or region).
- **Drop:** law-enforcement FTE (73.6% missing), raw teacher and counselor FTE, `LEA_SCHOOLS` (0.96 correlated with district enrollment).
- **Exclude as leakage:** seclusion components, rate, secluded-student counts, and all physical and mechanical restraint counts.

Data issues for the cleaning owners

- **High:** no analysis-ready flag and no filtered file, so rows with a missing outcome can be modeled by accident.
- **High:** IDEA suppression (-2) blanks idea_n even when one sex is known (4,741 schools); shares above 100% in 317 schools.
- **High:** law-enforcement FTE unusable (73.6% missing).
- **High:** confirm what -9 means for the seclusion outcome (not applicable vs not reported); it affects 1,806 schools and the prevalence.
- **Medium:** 135 schools have more instances than students; FTE outliers (483) are flagged but retained; the corrupted `COMBOKEY_stored` is still in the output; the small-enrollment flag is < 50 here but < 10 in 2017-18; the schema differs from 2017-18; grade band is not provided; race and a virtual indicator are not carried through.
- **Low:** one school with reserved codes in all Yes/No columns; 93 missing LEA enrollments.

The full list with row counts is in `data_issues_201516.csv`.

Open items

1. Supply 2021-22 (and 2020-21) cleaned files so the cross-year comparison can use the requested years.
2. Carry race through cleaning, then rerun the equity section.
3. Team decision: minimum enrollment of 30, whether the headline model includes state, and how -9 is treated if years are pooled.